Recurrence Relations, Markov Chains & Geometric Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Recurrence Relations, Markov Chains & Geometric Probability Practice Exercises

1 A sequence $\{a_n\}_{n \geq 0}$ satisfies $a_0 = 1$, $a_1 = 3$, and the linear recurrence $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \geq 2$. Compute the exact value of a_5.

a. 31

b. 65

c. 63 ✓

d. 59

The characteristic polynomial $t^2 - 3t + 2 = (t-1)(t-2)$ has distinct roots 1 and 2, so $a_n = A + B \cdot 2^n$ (equivalently $a_n - a_{n-1} = 2(a_{n-1} - a_{n-2})$). Iterating gives $a_2 = 7,\ a_3 = 15,\ a_4 = 31$, up to $a_5 = 63$.

2 A sequence $\{a_n\}_{n \geq 0}$ satisfies $a_0 = 2$, $a_1 = 4$, and the linear recurrence $a_n = 4a_{n-1} - 3a_{n-2}$ for $n \geq 2$. Compute the exact value of a_5.

a. 82

b. 244 ✓

c. 246

d. 238

The characteristic polynomial $t^2 - 4t + 3 = (t-1)(t-3)$ has distinct roots 1 and 3, so $a_n = A + B \cdot 3^n$ (equivalently $a_n - a_{n-1} = 3(a_{n-1} - a_{n-2})$). Iterating gives $a_2 = 10,\ a_3 = 28,\ a_4 = 82$, up to $a_5 = 244$.

3 A sequence $\{a_n\}_{n \geq 0}$ satisfies $a_0 = 3$, $a_1 = 6$, and the linear recurrence $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \geq 2$. Compute the exact value of a_5.

a. 96 ✓

b. 48

c. 99

d. 90

The characteristic polynomial $t^2 - 3t + 2 = (t-1)(t-2)$ has distinct roots 1 and 2, so $a_n = A + B \cdot 2^n$ (equivalently $a_n - a_{n-1} = 2(a_{n-1} - a_{n-2})$). Iterating gives $a_2 = 12,\ a_3 = 24,\ a_4 = 48$, up to $a_5 = 96$.

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4 A sequence $\{a_n\}_{n\ge 0}$ satisfies $a_0 = 4$, $a_1 = 7$, and the linear recurrence $a_n = 4a_{n-1} - 3a_{n-2}$ for $n \ge 2$. Compute the exact value of a_5.

a. 124

b. 370

c. 358

d. 367 ✓

The characteristic polynomial $t^2 - 4t + 3 = (t-1)(t-3)$ has distinct roots 1 and 3, so $a_n = A + B \cdot 3^n$ (equivalently $a_n - a_{n-1} = 3(a_{n-1} - a_{n-2})$). Iterating gives $a_2 = 16,\ a_3 = 43,\ a_4 = 124$, up to $a_5 = 367$.

5 Let $\{a_n\}_{n\ge 0}$ satisfy $a_0 = 1$, $a_1 = 4$, and $a_n = 5a_{n-1} - 6a_{n-2}$ $(n \ge 2)$. Find the exact value of a_5.

a. 146

b. 500

c. 454 ✓

d. 1006

The characteristic equation $t^2 - 5t + 6 = (t-2)(t-3) = 0$ yields $a_n = \alpha(2)^n + \beta(3)^n$. Equivalently, direct step-by-step evaluation gives $a_2 = 14,\ a_3 = 46,\ a_4 = 146$, and $a_5 = 454$.

6 Let $\{a_n\}_{n\ge 0}$ satisfy $a_0 = 2$, $a_1 = 5$, and $a_n = 5a_{n-1} - 6a_{n-2}$ $(n \ge 2)$. Find the exact value of a_5.

a. 97

b. 275 ✓

c. 310

d. 695

The characteristic equation $t^2 - 5t + 6 = (t-2)(t-3) = 0$ yields $a_n = \alpha(2)^n + \beta(3)^n$. Equivalently, direct step-by-step evaluation gives $a_2 = 13,\ a_3 = 35,\ a_4 = 97$, and $a_5 = 275$.

7 Let $\{a_n\}_{n\ge 0}$ satisfy $a_0 = 3$, $a_1 = 3$, and $a_n = a_{n-1} + 2a_{n-2}$ $(n \ge 2)$. Find the exact value of a_5.

a. 63 ✓

b. 33

c. 78

d. 3

The characteristic equation $t^2 - 1t - 2 = (t-2)(t+1) = 0$ yields $a_n = \alpha(2)^n + \beta(-1)^n$. Equivalently, direct step-by-step evaluation gives $a_2 = 9,\ a_3 = 15,\ a_4 = 33$, and $a_5 = 63$.

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8 Let $\{a_n\}_{n\geq 0}$ satisfy $a_0 = 1$, $a_1 = 5$, and $a_n = a_{n-1} + 2a_{n-2}$ $(n \geq 2)$. Find the exact value of a_5.

a. 31

b. 82

c. -3

d. 65 ✓

The characteristic equation $t^2 - 1t - 2 = (t-2)(t+1) = 0$ yields $a_n = \alpha(2)^n + \beta(-1)^n$. Equivalently, direct step-by-step evaluation gives $a_2 = 7,\ a_3 = 17,\ a_4 = 31$, and $a_5 = 65$.

9 A biased coin lands Heads with probability $p = \frac{1}{2}$ and Tails with probability $1 - p = \frac{1}{2}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 4

b. 10

c. 6 ✓

d. 7

Let e_r denote the expected flips to reach r consecutive Heads. To reach r Heads, we first reach $r-1$ Heads (taking e_{r-1} flips on average), then flip once more: with probability $\frac{1}{2}$ we finish, and with probability $\frac{1}{2}$ we reset to S_0. Thus $e_r = e_{r-1} + 1 + \frac{1}{2}e_r \Rightarrow e_r = 2(e_{r-1} + 1)$. Starting from $e_0 = 0$, $e_1 = 2$, $e_2 = 6$.

10 A biased coin lands Heads with probability $p = \frac{1}{3}$ and Tails with probability $1 - p = \frac{2}{3}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 9

b. 12 ✓

c. 21

d. 13

Let e_r denote the expected flips to reach r consecutive Heads. To reach r Heads, we first reach $r-1$ Heads (taking e_{r-1} flips on average), then flip once more: with probability $\frac{1}{3}$ we finish, and with probability $\frac{2}{3}$ we reset to S_0. Thus $e_r = e_{r-1} + 1 + \frac{2}{3}e_r \Rightarrow e_r = 3(e_{r-1} + 1)$. Starting from $e_0 = 0$, $e_1 = 3$, $e_2 = 12$.

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11 A biased coin lands Heads with probability $p = \frac{1}{4}$ and Tails with probability $1 - p = \frac{3}{4}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 20 ✓

b. 16

c. 36

d. 21

Let e_r denote the expected flips to reach r consecutive Heads. To reach r Heads, we first reach $r-1$ Heads (taking e_{r-1} flips on average), then flip once more: with probability $\frac{1}{4}$ we finish, and with probability $\frac{3}{4}$ we reset to S_0. Thus $e_r = e_{r-1} + 1 + \frac{3}{4}e_r \Rightarrow e_r = 4(e_{r-1} + 1)$. Starting from $e_0 = 0$, $e_1 = 4$, $e_2 = 20$.

12 A biased coin lands Heads with probability $p = \frac{1}{5}$ and Tails with probability $1 - p = \frac{4}{5}$. Using the Markov state transitions illustrated, find the exact expected number of flips required to observe 2 consecutive Heads (HH) for the first time.

a. 25

b. 55

c. 31

d. 30 ✓

Let e_r denote the expected flips to reach r consecutive Heads. To reach r Heads, we first reach $r-1$ Heads (taking e_{r-1} flips on average), then flip once more: with probability $\frac{1}{5}$ we finish, and with probability $\frac{4}{5}$ we reset to S_0. Thus $e_r = e_{r-1} + 1 + \frac{4}{5}e_r \Rightarrow e_r = 5(e_{r-1} + 1)$. Starting from $e_0 = 0$, $e_1 = 5$, $e_2 = 30$.

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